FTM iMCQ 05
Fall 2025

Histology, Genetics, Biochemistry

KEY file

FTM iMCQ 05

Histology

1. A 40-year-old woman comes to the physician because of a 3-year history of burning pain associated with meals, and a bitter taste sensation in her mouth. Physical examination shows tenderness on palpation of the epigastric region of her abdomen. Endoscopy is done and a tissue sample of the esophagus is obtained for biopsy. The biopsy shows the epithelium of her lower esophagus as a simple columnar epithelium with interspersed goblet cells. Which of the following cellular adaptations is most likely being observed?

A. Metaplasia
B. Hypertrophy
C. Hyperplasia
D. Involution
E. Atrophy

1. A 40-year-old woman comes to the physician because of a 3-year history of burning pain associated with meals, and a bitter taste sensation in her mouth. Physical examination shows tenderness on palpation of the epigastric region of her abdomen. Endoscopy is done and a tissue sample of the esophagus is obtained for biopsy. The biopsy shows the epithelium of her lower esophagus as a simple columnar epithelium with interspersed goblet cells. Which of the following cellular adaptations is most likely being observed?

A. Metaplasia
B. Hypertrophy
C. Hyperplasia
D. Involution
E. Atrophy

Define the term “metaplasia” and list examples. SOM.MK.I.BPM1.1.FTM.3.HCB.0841

2. A 61-year-old woman comes to the physician because of a 3-month history of blood in the stool and abdominal cramps. A digital rectal examination shows blood on the gloved finger. A colonoscopy is performed. During the procedure a mass is observed in the ascending colon. Microscopic examination of biopsy specimens of the mass shows rapidly dividing cells with abnormal arrangement and architectural disarray. Which of the following processes is most likely being described?

A. Involution
B. Atrophy
C. Metaplasia
D. Dysplasia
E. Hypertrophy

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A. Involution
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C. Metaplasia
✓ D. Dysplasia
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3. A 73-year-old woman comes to the physician for a routine examination. She has a history of right breast cancer for which she has completed treatment by radiation therapy 6 weeks ago. Some of the radiation doses were delivered to normal surrounding tissues, resulting in cell injury and the formation of a cystic mass. Tissue samples of the cyst were obtained by surgical resections. Western blot analysis of the cells from the cyst shows elevated levels of the Bcl-2-associated X (BAX) protein. Which of the following best identifies a function of the elevated protein?

A. Direct recruitment and activation of pro-caspase 9
B. Permeabilization of the mitochondrial membrane
C. Direct activation of executioner caspase 3
D. Marks apoptotic bodies for engulfment by macrophages
E. Degradation of cytoplasmic and nucleic proteins

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4. A 45-year-old woman comes to the physician because of a 3-month history of a “lump” on the right side of her neck. Physical examination shows a 3-cm firm and nontender mass on the right lateral neck. Examination of a biopsy specimen obtained from the mass shows an abundance of poorly differentiated cells, that have broken through the basement membrane. Genetic testing shows a mutation in the P53 gene, leading to a malignant neoplasm of the neck. The p53 gene is a tumor suppressor gene that halts the cell cycle and induces apoptosis, once there is insult to a cell. Which of the following best characterizes the type of cell death most likely to be affected by this mutation?

A. Lysis of plasma membrane
B. Intense inflammation
C. Damages to neighboring cells
D. Activation of caspase 9
E. Induced by acute injury

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5. A 55-year-old woman comes to the physician for a routine examination. She has a history of pancreatic cancer and has been enrolled in a chemotherapy treatment trial with a pro-apoptotic drug for the past 3 months. This drug is shown to cause cell death by transferring the membrane lipid phosphatidylserine from the inner leaflet to the outer leaflet of the cancer cell's membrane. Which of the following best identifies the function of this lipid transference in the type cell death most likely to occur?

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E. Degradation of cytoplasmic and nucleic proteins

Discuss the morphological changes associated with cell death by apoptosis.

FTM IMCQ5

Genetics

A 17-month-old boy is brought to the physician by his mother because of a history of progressive bowing of his legs. She says this is a family trait because she also has bowed legs and so does her father, paternal grandmother, and her paternal aunt. Physical examination shows widening of the wrists and genu varum on the limbs. In-depth sequencing of the gene encoding the vitamin D receptor on the X-chromosome is performed for the patient, and the results are shown. Sequencing identified a pathogenic mutation. The mother reports she is currently 20 weeks pregnant and has concerns about this child being affected by this disorder. Which of the following best represents the risk of her fetus being affected by this disorder?

A. 25%

B. 33%

C. 50%

D. 100%

E. 0% (zero)

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Reference sequence

   CATTACGGCGATTCGAGCTAACGT

Patient sequencing

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07 CATTACGGCGCTTCGAGCTAACGT
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09 CATTACGGCGCTTCGAGCTAACGT
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12 CATTACGGCGCTTCGAGCTAACGT

RESULTS

   ----------A-------------  = 0/12
   ----------C-------------  = 12/12
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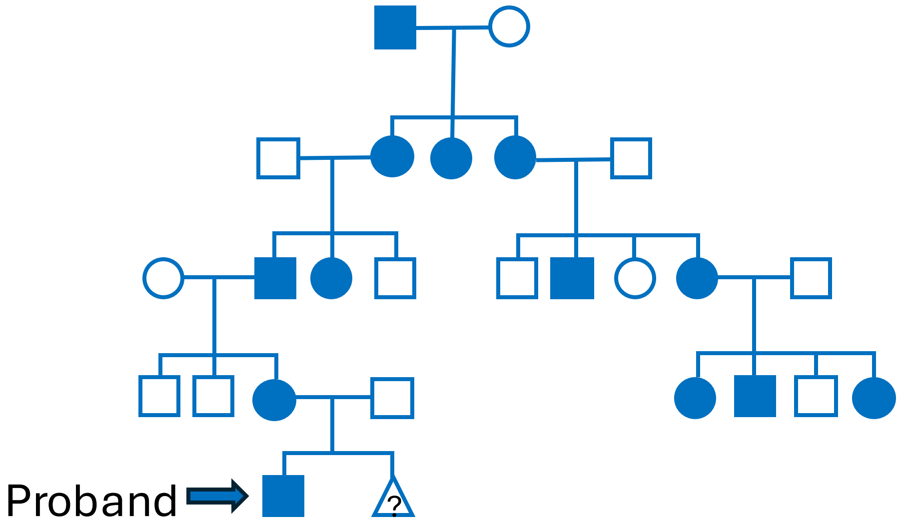

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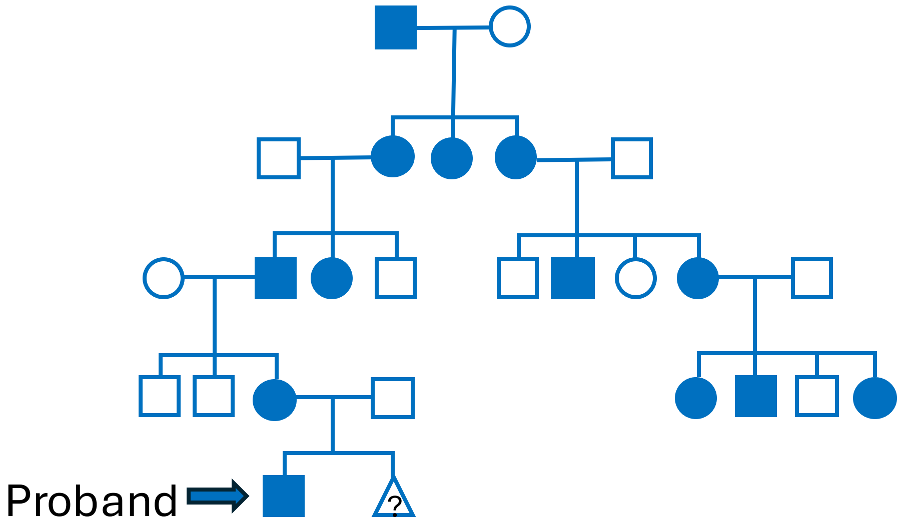

SOM.MK.III.BPM1.1.FTM.3.GNET.0081 Analyze examples of molecular diagnosis obtained from genetic and genomic laboratory analysis
SOM.MKIII.BPM1.1.FTM.3.GNET.0040 Propose how the basic risk calculations are performed for X-linked disorders

During a group session at a genetics clinic, several families meet and discover that their children all share similar craniofacial features, including downward-slanting palpebral fissures, micrognathia, and malformed ears. Genetic testing confirms a diagnosis of Treacher Collins syndrome in all children. However, further analysis reveals that mutations are found in different genes among the families: some in TCOF1, others in POLR1C, and others in POLR1D. Which of the following genetic principles best explains these finding?

A. Allelic heterogeneity
B. Variable expressivity
C. Locus heterogeneity
D. Pleiotropy
E. Incomplete penetrance

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SOM.MK.III.BPM1.1.FTM.3.GNET.0081Analyze examples of molecular diagnosis obtained from genetic and genomic laboratory analysis
SOM.MK.III.BPM1.1.FTM.3.GNET.0034Compare and contrast locus heterogeneity and allelic heterogeneity; give examples

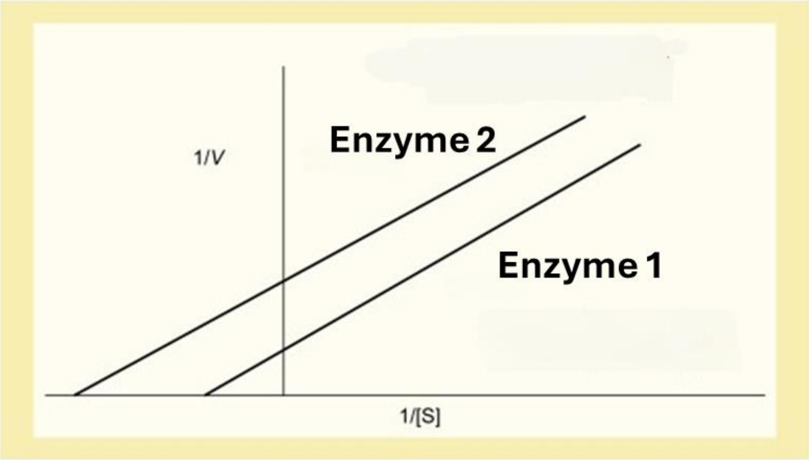
A. Allelic heterogeneity
B. Variable expressivity
C. Locus heterogeneity
D. Pleiotropy
E. Incomplete penetrance

FTM IMCQ5

Biochemistry

A 6-year-old boy is brought to the physician by his parents because of a 3-month history of fatigue and pale appearance. Laboratory studies show hemolytic anemia due to pyruvate kinase deficiency. Kinetic studies of the enzymes in the control (Enzyme 1) and the patient (Enzyme 2) are shown. Which of the following best represents the enzyme kinetics in the patient when compared to the control?

A. No change in Km and Vmax
B. Higher Vmax and lower Km
C. Higher Km and Vmax
D. Lower Vmax and higher Km
E. Lower Km and Vmax
F. Lower Vmax and lower substrate affinity



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SOM.MK.I.BPM1.1.FTM.3.BCHM.0111; Define Km and Vmax and discuss their significance with the help of a graph.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0115; Indicate X- and Y- intercepts in the Lineweaver-Burk plot

A. No change in Km and Vmax
B. Higher Vmax and lower Km
C. Higher Km and Vmax
D. Lower Vmax and higher Km
E. Lower Km and Vmax
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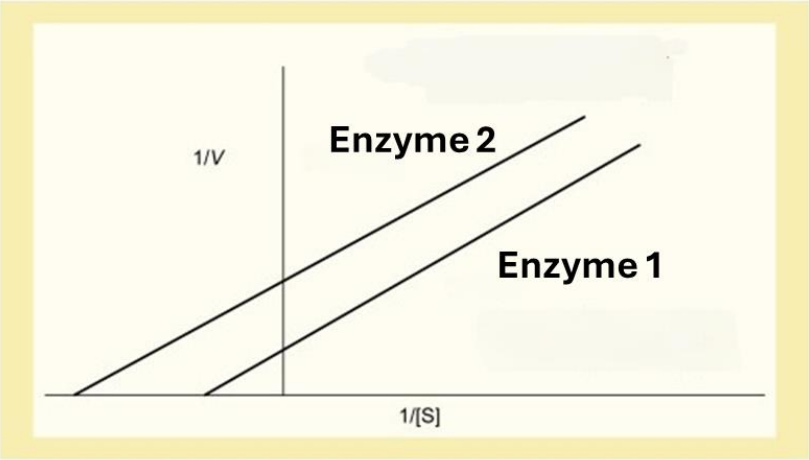

A 56-year-old man with a history of chronic hepatitis B infection comes to the physician because of progressive abdominal distension and unintentional weight loss. Physical examination shows scleral icterus, hepatomegaly, and shifting dullness. Laboratory studies show elevated liver enzymes and total bilirubin. Abdominal ultrasonography shows a hypervascular mass in the right hepatic lobe. Which of the following serum markers would most likely be elevated in this patient?

A. Total Creatine Kinase
B. Alpha-fetoprotein
C. Lipase
D. Lactate dehydrogenase
E. Acid phosphatase

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0142 Indicate specific serum injury markers for bone disease (ALP), prostate cancer (PSA) and live r cancer (alpha-fetoprotein).

A 5-year-old boy is brought to the physician by his parents for a follow up examination. He has a 2-month history of difficulty walking, climbing the stairs, or running. Physical examination shows proximal muscle atrophy. Photographs of the patient when asked to get up from the floor are shown. Muscle biopsy studies and nucleic acid amplification tests confirms the diagnosis of an inherited disorder. Which of the following best explains the molecular basis of the disorder in this patient?

A. Decreased dystrophin synthesis
B. In-frame deletion in dystrophin gene
C. Frameshift mutation in dystrophin gene
D. Skewed X-inactivation
E. Autosomal recessive inheritance

J.K. Mah, Neuropsychiatric Disease and Treatment, 2016

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0181
Describe the molecular basis for clinical features and diagnostic tests in Duchenne Muscular Dystrophy and Becker Muscular Dystrophy

Experiments are being conducted on the activity of a novel drug designed to modulate the activity of the LDL receptor metabolizing enzyme, PCSK9. Kinetic assays are performed in the presence and absence of the drug, and the resultant data are presented in the table. The data demonstrates that the drug is most likely acting as which of the following types of enzyme modulators?

	Velocity (μmol/min)	
Substrate concentration (μmol)	**Control**	**Presence of Drug**
1.0	1.6	0.4
3.0	4.8	2.3
5.0	6.1	2.9
10.0	10.1	4.1
20.0	11.8	7.5
30.0	12.2	12.0

A. Inducer of the enzyme
B. Allosteric activator
C. Allosteric inhibitor
D. Competitive inhibitor
E. Non-competitive inhibitor

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0111; Define Km and Vmax and discuss their significance with the help of a graph.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0113; Compare and contrast the mode of action and kinetics of reversible competitive and noncompetitive inhibitors.

A 23-year-old man comes to the emergency department because of a 6-hour history of worsening and severe abdominal pain. He describes the pain as dull, severe and radiating to the back. He also says the pain is less when he is sitting. He says the pain occurs after binge drinking alcohol with his friends. He says he never experienced this pain before, and this is the first occurrence. His pulse is 100/min, respirations are 25/min, and blood pressure is 110/80 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 96%. Physical examination shows jaundice (yellowing of his sclerae) and breath smells of alcohol. Laboratory studies are shown. Which of the following best explains the lab findings and presentation in this patient?

Marker	**Patient**	**Normal Range**
Lipase	500 U/L	0-70 U/L
Amylase	450 U/L	0-90 U/L
GGT	150 IU/L	0-55 IU/L
Bilirubin	60 umol/L	3-22umol/L
Alkaline Phosphatase	150 IU/L	100 IU/L
AST	60 IU/L	10-40 IU/L
ALT	65 IU/L	0-50 IU/L

A. Alcohol related pancreatitis
B. Viral hepatitis
C. Gallstone pancreatitis
D. Alcohol induced hepatitis
E. Gastritis

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B. Viral hepatitis
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SOM.MK.I.BPM1.1.FTM.3.BCHM.0140 Describe the application of ALT, AST, ALP and GGT as markers of hepatocellular disease, biliary disease and alcohol liver disease.

A 22-year-old man comes to the physician because of a 1-day history of painful, prolonged erection without any sexual stimulation. Three months ago, he was diagnosed with schizophrenia and was prescribed a tri-drug treatment including risperidone. This drug is an alpha1-adrenergic receptor antagonist. Priapism is a known adverse effect of this drug. Which of the following enzyme activities is most likely decreased in the target cells of this patient following administration of the drug?

A. Guanylate cyclase
B. Adenylate cyclase
C. Phosphodiesterase
D. Protein kinase A
E. Phospholipase C
F. Protein kinase G

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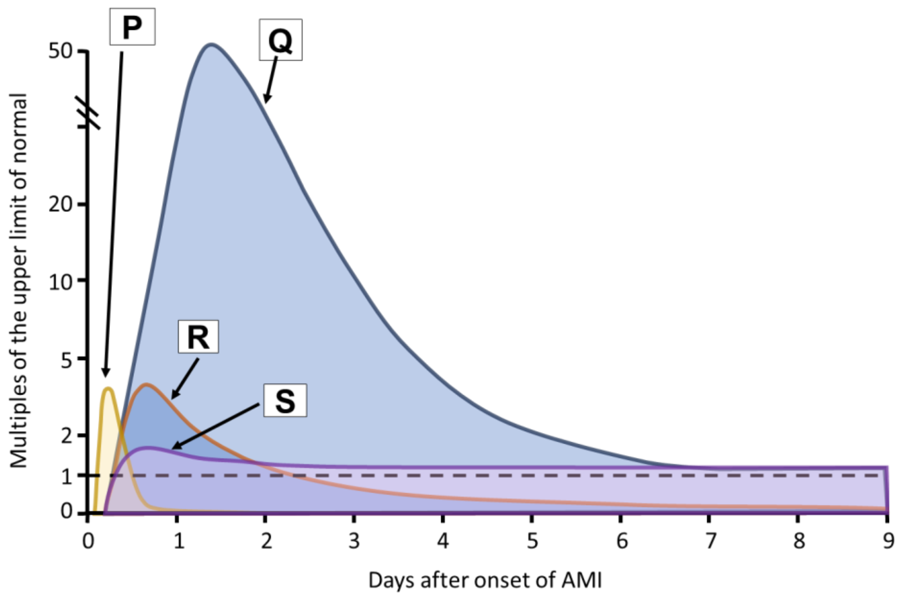
SOM.MK.I.BPM1.1.FTM.3.BCHM.0126 Describe the two major second messenger systems (adenylate cyclase and phosphoinositide systems) associated with G-proteins and signal termination.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0135 Predict the changes when an agonist or antagonist of a specific signaling pathway is administered.

A 55-year-old man with familial hyperlipidemia, a BMI of 28.7 kg/m^2 is admitted to the emergency department after he collapsed with severe chest pain while out walking in the park. Results of an ECG and serum studies confirm a diagnosis of myocardial infarction. He was prescribed anticoagulant therapy and due to his medical history, he was admitted to the hospital Coronary Care Unit. Blood samples were analyzed every 24 hours for a series of cardiac biomarkers. The data from these samples were plotted on the graph. Which of the following options best identifies biomarkers **P** and **Q**?

	P	Q
A.	CK-MB	Myoglobin
B.	Myoglobin	Troponin
C.	Lactate dehydrogenase	Aspartate aminotransferase
D.	Troponin	CK-MB
E.	CK-MB	Alkaline phosphatase



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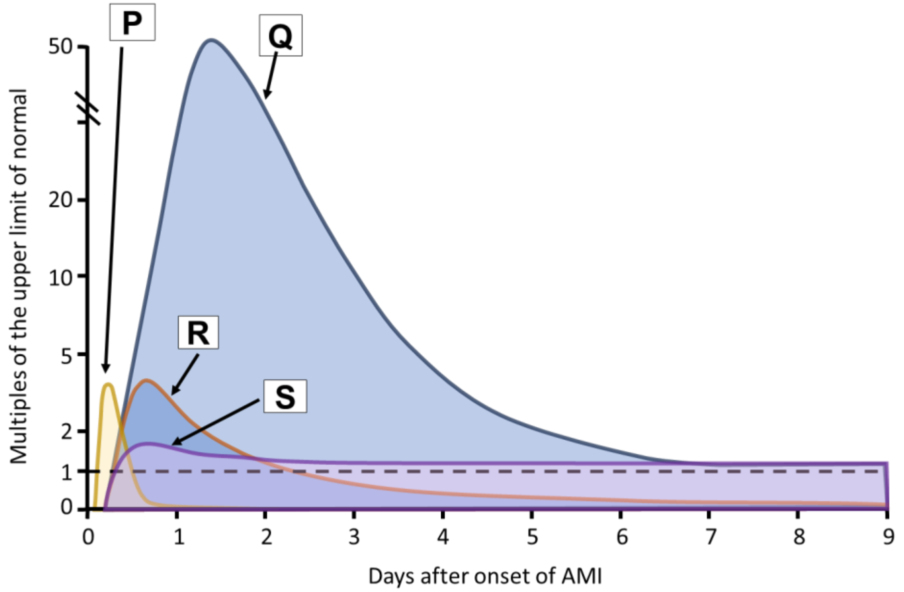

SOM.MK.I.BPM1.1.FTM.3.BCHM.0138 Discuss utility of serum injury markers following myocardial infarction: Myoglobin, CK-MB/total CK ratio, serum cardiac troponins I and T

Investigators are studying the characteristics of the enzyme fructose 1,6-bisphosphatase which catalyzes one of the regulated steps of gluconeogenesis. This enzyme is regulated by allosteric inhibition by fructose 2,6-bisphosphate. Which of the following is most likely observed in the presence of a high concentration of this modulator?

A. A left shift of the curve
B. A smaller $K_{0.5}$ for substrate
C. Increased degradation of enzyme
D. Lower substrate affinity
E. Decreased enzyme synthesis

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0109 Describe allosteric enzymes and their regulation related to changes of $K_{0.5}$ or Vmax. Discuss in general the regulation of metabolic pathways related to feedback inhibition and feed-forward activation.

A 34-year-old man comes to the emergency department because of a 2-day history of profuse watery diarrhea after returning from a trip abroad. Physical examination shows he is dehydrated with dry mucous membranes and poor skin turgor. Laboratory studies show hypokalemia and metabolic acidosis. Stool cultures grow *Vibrio cholerae*. Which of the following mechanisms best explains the effect of the toxin produced by this microorganism on intestinal epithelial cells?

A. Activation of Gi-protein
B. Irreversible activation of adenylyl cyclase
C. Locking of Gs-alpha subunit in the inactive form
D. Prevention of GTP binding to Gq subunit
E. Rapid hydrolysis of G-protein bound GTP

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0127; Explain how cholera toxin modifies Gs and how pertussis toxin modifies Gi. Predict the effects of these modifications in the respective cells.